

ST QUENTIN ROUPY AP, France

WMO: 070610

Lat: 49.8183N

Lon: 3.2061E

Elev: 101

StdP: 100.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-3.8	-9.0	1.8	-4.4	-6.6	2.2	-3.0	13.7	9.5	12.2	9.0	3.4	40	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.5	20.2	27.4	19.5	25.4	18.6	21.2	27.6	20.2	25.8	19.2	24.1	3.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.0	24.0	18.1	13.2	22.4	17.3	12.5	21.5	62.0	27.8	58.2	26.0	55.1	24.4	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
10.8	9.4	8.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-8.2	33.6	3.3	2.3	-10.5	35.2	-12.4	36.6	-14.3	37.9	-16.6	39.6		
			-8.6	23.3	3.2	1.0	-10.9	24.1	-12.7	24.7	-14.5	25.3	-16.8	26.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	3.4	4.2	6.9	9.8	13.2	16.0	18.3	18.0	15.1	11.4	7.0	4.1
	DBStd	6.29	4.10	3.66	3.33	3.50	3.31	3.11	3.15	2.88	2.83	3.35	3.54	3.96
	HDD10.0	841	207	164	104	46	8	0	0	0	1	24	102	186
	HDD18.3	2901	464	397	353	257	164	84	40	40	105	214	341	442
	CDD10.0	1078	1	1	9	39	106	181	258	249	152	68	11	2
	CDD18.3	96	0	0	0	0	4	14	40	31	7	0	0	0
	CDH23.3	945	0	0	0	6	40	150	394	301	52	2	0	0
	CDH26.7	256	0	0	0	0	4	35	115	94	7	0	0	0
Wind	WSAvg	4.1	4.8	4.7	4.6	4.1	3.8	3.5	3.5	3.4	3.6	4.1	4.3	4.6
Precipitation	PrecAvg	722	59	53	52	47	59	64	62	70	54	64	64	74
	PrecMax	976	161	112	170	126	127	121	121	169	115	137	132	187
	PrecMin	538	7	13	4	3	3	27	17	7	10	23	16	27
	PrecStd	109	31	28	33	28	32	31	26	37	28	28	28	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.0	18.9	24.1	27.0	30.4	32.7	32.9	27.6	22.5	16.5	13.5
		MCWB	11.4	10.2	12.2	15.5	20.0	21.4	21.1	21.2	19.4	17.4	14.5	11.9
	2%	DB	11.7	11.8	16.2	21.2	24.3	27.1	29.5	29.5	24.7	19.7	14.5	12.1
		MCWB	10.4	9.7	11.5	14.5	17.5	20.1	20.0	20.1	18.2	16.2	12.9	10.9
	5%	DB	10.2	10.6	13.9	18.2	22.0	24.9	27.4	26.8	22.2	17.7	13.2	11.0
		MCWB	9.0	9.0	10.4	12.7	16.4	18.9	19.3	19.1	16.9	15.0	11.9	10.0
	10%	DB	8.9	9.5	12.2	16.0	19.7	22.6	25.3	24.3	20.3	16.2	12.1	9.7
		MCWB	7.9	8.1	9.7	11.6	15.0	17.5	18.6	17.8	16.0	14.2	11.0	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.1	13.3	17.1	20.7	22.6	22.2	22.6	20.3	18.0	15.0	12.3
		MCDB	12.7	12.8	16.5	22.9	26.0	28.4	29.3	30.9	25.6	21.2	16.1	13.1
	2%	WB	10.5	10.0	12.2	14.8	18.4	20.6	21.1	20.9	18.9	16.6	13.2	11.1
		MCDB	11.5	11.2	15.1	19.6	22.9	25.5	27.7	27.6	23.0	18.9	14.2	12.0
	5%	WB	9.1	9.1	11.1	13.3	16.8	19.4	20.1	19.7	17.7	15.6	12.1	10.0
		MCDB	10.0	10.4	13.4	17.4	20.9	24.0	26.0	25.2	21.1	17.4	13.1	10.9
	10%	WB	8.0	8.2	10.0	12.1	15.5	18.1	19.0	18.6	16.8	14.5	11.1	8.9
		MCDB	8.9	9.4	11.9	15.3	19.0	22.0	24.0	23.0	19.6	16.0	12.0	9.7
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.7	7.7	9.5	9.4	10.1	10.6	10.4	9.3	7.5	5.4	4.5
		MCDBR	5.3	7.2	11.0	12.9	13.2	13.8	14.0	14.4	11.9	9.0	6.0	5.0
		MCWBR	4.8	5.6	7.1	7.3	7.4	7.5	6.1	6.4	6.4	5.8	4.8	4.6
	5% WB	MCDBR	5.2	6.5	9.2	11.3	11.6	12.6	12.2	12.6	10.3	8.2	5.6	5.0
		MCWBR	4.9	5.4	6.3	6.8	7.0	7.3	5.8	6.2	5.9	5.6	4.7	4.6
Clear-Sky Solar Irradiance	taub	0.323	0.343	0.377	0.406	0.415	0.414	0.414	0.414	0.404	0.377	0.361	0.333	0.318
	taud	2.413	2.375	2.269	2.203	2.216	2.246	2.245	2.297	2.359	2.407	2.417	2.381	
	Ebn at Noon	719	793	821	836	844	846	839	829	815	760	695	662	
	Edh at Noon	64	84	110	131	135	132	130	118	100	81	64	58	
All-Sky Solar Radiation	RadAvg	0.86	1.58	2.75	4.30	5.04	5.55	5.27	4.51	3.48	2.04	1.01	0.69	
	RadStd	0.10	0.27	0.35	0.49	0.49	0.52	0.49	0.45	0.28	0.17	0.10	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page